

Longest Common Subsequence

The Longest Common Subsequence (LCS) problem finds the longest subsequence common to two sequences. A subsequence is a sequence that appears in the same relative order, but not necessarily contiguous.

Example: $X = \text{"ABCBDAB"}$, $Y = \text{"BDCABA"}$

For a smaller subproblem to illustrate the table, let $X = \text{"ABCBD"}$ and $Y = \text{"BDCAB"}$.

To solve this with Dynamic Programming, we define $dp[i][j]$ as the length of the LCS of the first i characters of X and the first j characters of Y .

The DP recurrence is:

If $X_i = Y_j$:

$$dp[i][j] = 1 + dp[i-1][j-1]$$

Else:

$$dp[i][j] = \max(dp[i-1][j], dp[i][j-1])$$

Base case: $dp[0][j] = 0$ and $dp[i][0] = 0$ for all i, j .

Here is the DP table for $X = \text{"ABCBD"}$ and $Y = \text{"BDCAB"}$:

		B	D	C	A	
		B	D	C	A	B
		0	0	0	0	0
A	A	0	0	0	1	1
B	B	0	1	1	1	2
C	C	0	1	1	2	2
B	B	0	1	1	2	3
D	D	0	1	2	2	3

To find the actual LCS string, we backtrack from the bottom-right cell, $dp[5][5] = 3$.

Path:

- (5,5): $X = \text{"D"}, Y = \text{"B"} \rightarrow D \neq B$. $dp[4][5] = 3$, $dp[5][4] = 2$. Move $\downarrow$ to (4,5).
- (4,5): $X = \text{"B"}, Y = \text{"B"} \rightarrow \text{Match!}$ Include "B". Move $\swarrow$ to (3,4).
- (3,4): $X = \text{"C"}, Y = \text{"A"} \rightarrow C \neq A$. $dp[2][4] = 1$, $dp[3][3] = 2$. Move $\downarrow$ to (3,3).
- (3,3): $X = \text{"C"}, Y = \text{"C"} \rightarrow \text{Match!}$ Include "C". Move $\swarrow$ to (2,2).
- (2,2): $X = \text{"B"}, Y = \text{"D"} \rightarrow B \neq D$. $dp[1][2] = 0$, $dp[2][1] = 1$. Move $\leftarrow$ to (2,1).
- (2,1): $X = \text{"B"}, Y = \text{"B"} \rightarrow \text{Match!}$ Include "B". Move $\swarrow$ to (1,0).
- (1,0): Value is 0. Stop.

The collected characters in reverse order are "B", "C", "B". Reversing them gives the LCS: "BCB".

Key Takeaways: LCS is solved using a 2D DP table where $dp[i][j]$ depends on